Class X Maths (Standard) Sample Paper – Set 3

Max. Marks: 80 **Time Allowed: 3 hours**

General Instructions:

1. This Question Paper has 5 Sections A, B, C, D and E.

2. In Section A, Questions no. 1-18 are multiple choice questions (MCQs) and questions no. 19 and 20 are Assertion- Reason based questions of 1 mark each.

3. Section B has 5 questions carrying 02 marks each.

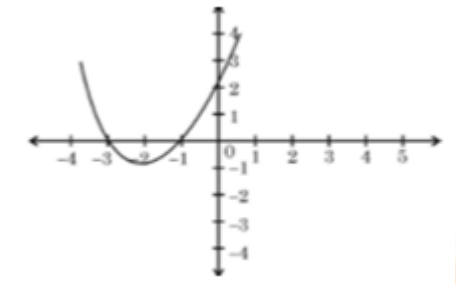
4. Section C has 6 questions carrying 03 marks each.

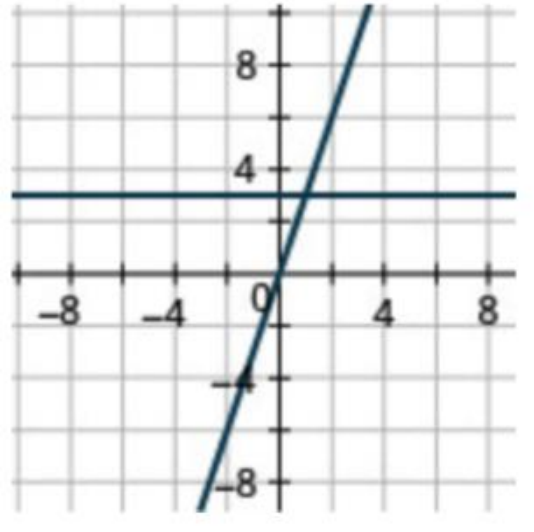
5. Section D has 4 questions carrying 05 marks each. Maximum Marks: 80

6. Section E has 3 case based integrated units of assessment (04 marks each) with sub- parts of the values of 1, 1 and 2 marks each respectively.

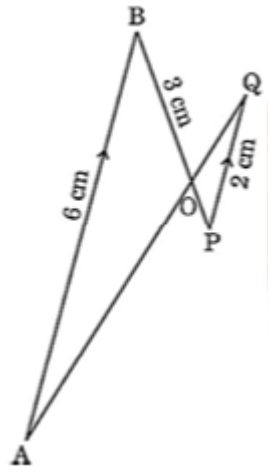
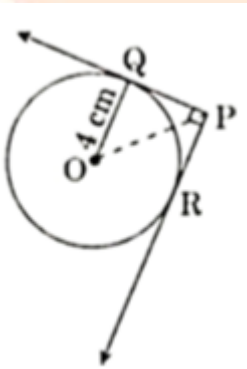
7. All Questions are compulsory. However, an internal choice in 2 Qs of 5 marks, 2 Qs of 3 marks and 2 Questions of 2 marks has been provided. An internal choice has been provided in the 2marks questions of Section E.

8. Draw neat figures wherever required. Take $\pi = \frac{22}{7}$ wherever required if not stated.

Section A		20
Q. No.	**Questions**	**Marks**
1.	The number $\frac{\sqrt{5}+\sqrt{2}}{\sqrt{5}-\sqrt{2}}$ is a) an irrational number b) an integer c) not a real number d) a rational number	1
2.	In the figure, the graph of the polynomial p(x) is given. The number of zeroes of the polynomial is:  a) 2 b) 1 c) 0 d) 3	1

3.	The number of solutions of two linear equations representing intersecting lines is/are  a) 1 b) 2 c) 0 d) ∞	1
4.	The value of λ for which $(x^2 + 4x + \lambda)$ is a perfect square, is a) 1 b) 16 c) 4 d) 9	1

5.	If the sum of n terms of an A.P. is $3n^2 + 5n$ then which of its term is 164? a) 27th b) none of these. c) 28th d) 26th	1
6.	The diameter of a circle is of length 6 cm. If one end of the diameter is (-4, 0), the other end on x-axis is at: a) (6, 0) b) (0, 2) c) (4, 0) d) (2, 0)	1
7.	If the point C(k, 4) divides the join of the points A(2, 6) and B(5,1) in the ratio 2:3 then the value of k is a) 16 b) $\frac{28}{5}$ c) $\frac{8}{5}$ d) $\frac{16}{5}$	1

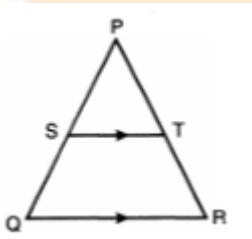
8.	In the given figure, AB || PQ. If AB = 6 cm, PQ = 2 cm and OB = 3 cm, then the length of OP is:  a) 1 cm b) 4 cm c) 9 cm d) 3 cm	1
9.	In the given figure, from an external point P, two tangents PQ and PR are drawn to a circle of radius 4 cm with centre O. If $\angle QPR = 90^0$, then length of PQ is:  	1

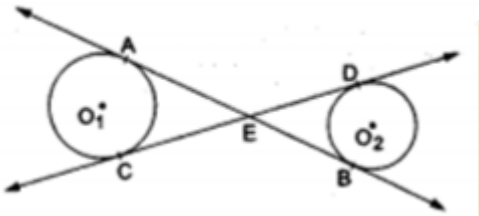
	a) 3 cm b) 4 cm c) 2 cm d) 2 $\sqrt{2}$ cm	
10.	From an external point Q, the length of tangent to a circle is 12 cm and the distance of Q from the centre of circle is 13 cm. The radius of circle (in cm) is a) 10 b) 5 c) 7 d) 12	1
11.	If $x = a\cos\theta$ and $y = b\sin\theta$, then the value of $b^2x^2 + a^2y^2$ is a) a + b b) a^2b^2 c) a - b d) ab	1
12.	$(3\sin^2 30^\circ - 4\cos^2 60^\circ)$ is equal to: a) $-\frac{1}{4}$	1

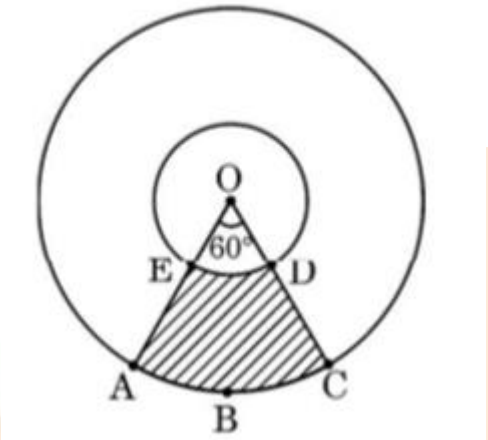
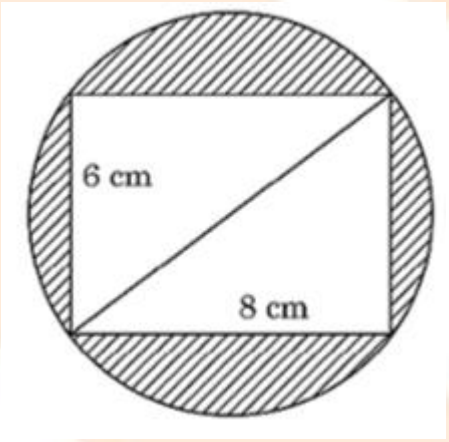
	b) $\frac{5}{4}$ c) $-\frac{3}{4}$ d) $-\frac{9}{4}$	
13.	The top of a broken tree has its top end touching the ground at a distance 15 m from the bottom, the angle made by the broken end with the ground is 30^0. Then length of broken part is a) 10 m b) $10\sqrt{3}$ m c) $\sqrt{3}$ m d) $5\sqrt{3}$ m	1
14.	The area of the sector of a circle with radius 6 cm which subtends an angle of 60° at the centre of the circle is: a) $\frac{152}{7}cm^2$ b) $\frac{132}{7}cm^2$ c) $\frac{142}{7}cm^2$ d) $\frac{122}{7}cm^2$	1

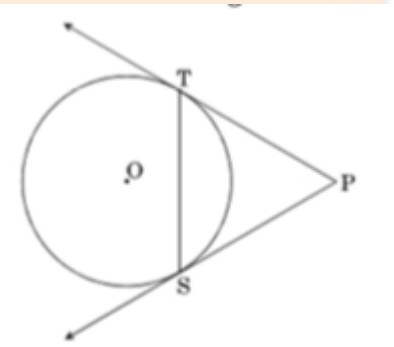
15.	A piece of wire 20cm long is bent into the form of an arc of a circle subtending an angle of 60^0 at its centre. The radius of the circle is a) $\frac{20}{6+\pi}cm$ b) $\frac{30}{6+\pi}cm$ c) $\frac{60}{\pi}cm$ d) $\frac{15}{6+\pi}cm$	1
16.	A bag contains 100 cards numbered 1 to 100. A card is drawn at random from the bag. What is the probability that the number on the card is a perfect cube? a) $\frac{1}{20}$ b) $\frac{1}{25}$ c) $\frac{7}{100}$ d) $\frac{3}{50}$	1
17.	If a letter of English alphabet is chosen at random, then the probability of this letter to be a consonant is:	1

	a) $\frac{11}{13}$ b) $\frac{10}{13}$ c) $\frac{21}{26}$ d) $\frac{5}{26}$	
18.	If the difference of mode and median of a data is 24, then the difference of median and mean of the same data is: a) 8 b) 12 c) 34 d) 24	1
19.	**Assertion (A):** In a solid hemisphere of radius 10 cm, a right cone of same radius is removed out. The surface area of the remaining solid is 570.74 cm^2 [Take $\pi = 3.14$ and $\sqrt{2} = 1.4$] **Reason (R):** Expression used here to calculate Surface area of remaining solid = Curved surface area of hemisphere + Curved surface area of cone a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.	1

	c) A is true but R is false. d) A is false but R is true	
20.	**Assertion (A):** The 11th term of an AP is 7, 9, 11, 13 is 67. **Reason (R):** If s_n is the sum of first n terms of an AP then its nth term an is given by $a_n = s_n + s_{n-1}$. a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A. c) A is true but R is false. d) A is false but R is true	1
	Section B	**10**
21.	Prove that $5\sqrt{2}$ is irrational.	2
22.	In the given figure, in a triangle ΔPQR, $ST \parallel QR$ and $\frac{PS}{SQ} = \frac{3}{5}, PR = 28\text{cm}$, find PT.  	2

23.	In the given figure, common tangents AB and CD to the two circles with centres O_1 and O_2 intersect at E. Prove that AB = CD.  	2
24.	Find the value of: $\sin 30^\circ \cdot \cos 60^\circ + \cos 30^\circ \cdot \sin 60^\circ$. Is it equal to $\sin 90^\circ$ or $\cos 90^\circ$? **OR** If $a\cos\theta + b\sin\theta = m$ and $a\sin\theta - b\cos\theta = n$, prove that: $a^2 + b^2 = m^2 + n^2$.	2
25.	In the given figure, two concentric circles with centre O are shown. Radii of the circles are 2 cm and 5 cm respectively. Find the area of the shaded region.	2

	 **OR** Reeti prepares a Rakhi for her brother Ronit. The Rakhi consists of a rectangle of length 8 cm and breadth 6 cm inscribed in a circle as shown in the figure. Find the area of the shaded region. (Use π = 3 14) 	
	Section C	**18**
26.	Find the greatest number that will divide 445, 572 and 699 leaving remainders 4, 5 and 6 respectively.	3

27.	The angle of elevation of a jet plane from a point A on the ground is 60°. After a flight of 30 seconds, the angle of elevation changes to 30°. If the jet plane is flying at a constant height of $3600\sqrt{3}$ m find the speed of the jet plane.	3
28.	If the last term of an A.P. of 30 terms is 119 and the 8th term from the end (towards the first term) is 91, then find the common difference of the A.P. Hence, find the sum of all the terms of the A.P. **OR** The sum of 5th and 9th terms of an A.P. is 72 and the sum of 7th and 12th terms is 97. Find the A.P.	3
29.	If radii of the two circles are equal, prove that AB = CD where AB and CD are common tangents. **OR** In the given figure, PT and PS are tangents to a circle with centre O, from a point P, such that PT = 4 cm and TPS = 60°. Find the length of the chord TS. Also, find the radius of the circle.  	3
30.	Prove that $\frac{1+\tan^2 A}{1+\cot^2 A} = \left(\frac{1-\tan A}{1-\cot A}\right)^2 = \tan^2 A.$	3

31. Find the mean of the following frequency distribution: — 3

Class	Frequency
0-10	12
10-20	18
20-30	27
30-40	20
40-50	17
50-60	6

Section D — **20**

32. A cottage industry produces a certain number of pottery articles in a day. It was observed on a particular day that cost of production of each article (in rupees) was 3 more than twice the number of articles produced on that day. If, the total cost of production on that day was ₹ 90, find the number of articles produced and the cost of each article. — 5

OR

If (- 5) is a root of the quadratic equation $2x^2 + px + 15 = 0$ and the quadratic equation $p(x^2 + x) + k = 0$ has equal roots, then find the values of p and k.

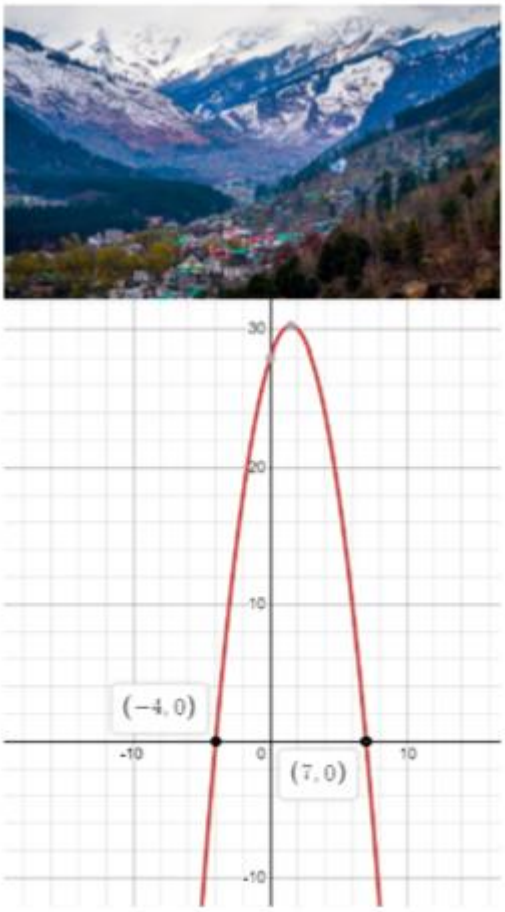
<table>
<tr><td>33.</td><td>In a trapezium ABCD, $AB \| DC$ and DC = 2AB. $EF \| AB$, where E and F lie on BC and AD respectively such that $\frac{BE}{EC} = \frac{4}{3}$. Diagonal DB intersects EF at G. Prove that 7EF = 11AB.</td><td>5</td></tr>
<tr><td>34.</td><td>A solid is in the shape of a right-circular cone surmounted on a hemisphere, the radius of each of them being 7 cm and the height of the cone is equal to its diameter. Find the volume of the solid.

OR

A tent is in the shape of a right circular cylinder up to a height of 3 m and then a right circular cone, with a maximum height of 13.5 m above the ground. Calculate the cost of painting the inner side of the tent at the rate of ₹ 2 per square metre, if the radius of the base is 14 m.</td><td>5</td></tr>
<tr><td>35.</td><td>The following table shows the ages of the patients admitted in a hospital during a month:

<table><tr><td>Age(in years)</td><td>6-15</td><td>16-25</td><td>26-35</td><td>36-45</td><td>46-55</td><td>56-65</td></tr><tr><td>Number of Patients</td><td>6</td><td>11</td><td>21</td><td>23</td><td>14</td><td>5</td></tr></table>Find the mode and the mean of the data given above. Compare and interpret the two measures of central tendency.</td><td>5</td></tr>
<tr><td colspan="2">Section E</td><td>12</td></tr>
<tr><td>36.</td><td>Read the following text carefully and answer the questions that follow:</td><td>4</td></tr>
</table>

	Two schools P and Q decided to award prizes to their students for two games of Hockey ₹ x per student and Cricket ₹ y per student. School P decided to award a total of ₹ 9,500 for the two games to 5 and 4 students respectively; while school Q decided to award ₹ 7,370 for the two games to 4 and 3 students respectively.  **i.** Represent the following information algebraically (in terms of x and y). **ii.** What is the prize amount for hockey? **iii.** Prize amount on which game is more and by how much? **OR** What will be the total prize amount if there are 2 students each from two games?	
37.	**Read the following text carefully and answer the questions that follow:** Two friends Govind and Pawan decided to go for a trekking. During summer vacation, they went to Panchmarhi. While trekking they observed that the trekking path is in the shape of a parabola. The mathematical representation of the track is shown in the graph.	4



i. What are the zeroes of the polynomial whose graph is given?

ii. What will be the expression of the given polynomial p(x)?

iii. What is the product of the zeroes of the polynomial which represents the parabola?

OR

In the standard form of quadratic polynomial, $ax^2 + bx + c$, whar are a, b, and c?

38.	**Read the following text carefully and answer the questions that follow:** Using Cartesian Coordinates we mark a point on a graph by how far along and how far up it is. The left-right (horizontal) direction is commonly called X-axis. The up-down (vertical) direction is commonly called Y-axis. When we include negative values, the x and y axes divide the space up into 4 pieces. **Read the information given above and below:** Two friends Veena and Arun work in the same office in Delhi. In the Christmas vacations, both decided to go their hometowns represented by Town A and Town B respectively in the figure given below. Town A and Town B are connected by trains from the same station C (in the given figure) in Delhi.  **i.** Who will travel more distance to reach their home? **ii.** Find the location of the station. **iii.** Find in which ratio Y-axis divide Town B and Station.	4

	OR Find the distance between Town A and Town B.	

Answer Key

Section A

1. a) an irrational number

2. a) 2

3. a) 1

4. c) 4

5. a) 27th

6. d) (2, 0)

7. d) $\frac{16}{5}$

8. a) 1 cm

9. b) 4 cm

10. b) 5

11. b) a^2b^2

12. a) $-\frac{1}{4}$

13. b) $10\sqrt{3}$ m

14. b) $\frac{132}{7}cm^2$

15. c) $\frac{60}{\pi}cm$

16. b) $\frac{1}{25}$

17. c) $\frac{21}{26}$

18. b) 12

19. d) A is false but R is true.

20. c) A is true but R is false.

Section B

21. Let us assume that $5\sqrt{2}$ is rational. Then, there exist positive co-primes a and b such that

$5\sqrt{2} = \frac{a}{b}$.

$\sqrt{2} = \frac{a}{b} - 5$.

$\sqrt{2} = \frac{a-5b}{b}$.

As a - 5b and b are integers,

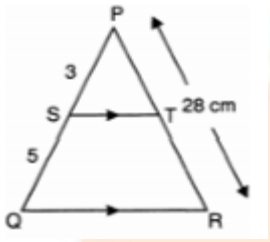
So, $\frac{a-5b}{b}$ is a rational number.

But $\sqrt{2}$ is not a rational number.

Since a rational number cannot be equal to an irrational number, our assumption that $5\sqrt{2}$ is rational is wrong.

Hence, $5\sqrt{2}$ is irrational.

22. According to the question,



$ST \parallel QR$

$\therefore \frac{PS}{PQ} = \frac{PT}{PR}$ (By BPT) Given, $\frac{PS}{SQ} = \frac{3}{5}$ and PR = 28 cm,

PQ = 3 + 5 = 8.

Or, $\frac{PS}{PQ} = \frac{PT}{PR}$.

Or, $\frac{3}{8} = \frac{PT}{28}$.

$\therefore PT = \frac{3 \times 28}{8} = 10.5cm.$

23. We know that tangent segments to a circle from the same external point are congruent.

So, EA=EC for the circle having centre O_1

And, ED=EB for the circle having centre O_2

Now, Adding ED on both sides in EA = EC, we get

EA+ED=EC+ED

⇒EA+EB=EC+ED

⇒AB=CD

24. $\sin 30^\circ \cdot \cos 60^\circ + \cos 30^\circ \cdot \sin 60^\circ$

$= \frac{1}{2} \times \frac{1}{2} + \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2}$

$= \frac{1}{4} + \frac{3}{4}$

$= \frac{4}{4}$

$= 1 = \sin 90^\circ = \cos 90^\circ.$

OR

Given, R.H.S $= m^2 + n^2$

$= (a\cos\theta + b\sin\theta)^2 + (a\sin\theta - b\cos\theta)^2$ [since, $m = a\cos\theta + b\sin\theta$ and $n = a\sin\theta - b\cos\theta$]

$= (a^2\cos^2\theta + b^2\sin^2\theta + 2ab\cos\theta\sin\theta) + (a^2\sin^2\theta + b^2\cos^2\theta - 2ab\sin\theta\cos\theta)$ [using $(a \pm b)^2 = a^2 + b^2 \pm 2ab$]

$= a^2(\cos^2\theta + \sin^2\theta) + b^2(\sin^2\theta + \cos^2\theta) = a^2 + b^2$ [since, $\sin^2\theta + \cos^2\theta = 1$].

Therefore, $m^2 + n^2 = a^2 + b^2$.

Hence proved.

25. Area of sector OABC:

$= \frac{\pi \times 5^2 \times 60^\circ}{360^\circ} = \frac{25\pi}{6} \text{cm}^2.$

Area of sector OED:

$= \frac{\pi \times 2^2 \times 60^\circ}{360^\circ} = \frac{4\pi}{6} \text{cm}^2.$

Area of shaded region:

$= \frac{25\pi}{6} - \frac{4\pi}{6} = \frac{21\pi}{6} \times \frac{22}{7} = 11\text{cm}^2.$

OR

Diagonal of rectangle:

$= \sqrt{6^2 + 8^2} = 10.$

$\therefore$ Radius of circle $r = \frac{10}{2} = 5.$

Area of circle: $= 3.14 \times 5 \times 5 = 78.5\text{cm}^2.$

Area of rectangle: $= 6 \times 8 = 48\text{cm}^2.$

Area of shaded region: $= 78.5 - 48 = 30.5\text{cm}^2.$

$\therefore$ Area of shaded region is $30.5\text{cm}^2.$

Section C

26. We have to find the greatest number that divides 445, 572 and 699 and leaves remainders of 4, 5 and 6 respectively. This means when the number divides 445, 572 and 699, it leaves remainders 4, 5 and 6. It means that

445 - 4 = 441,

572 - 5 = 567

and 699 - 6 = 693 are completely divisible by the required number.

For the highest number which divides the above numbers we need to calculate HCF of 441, 567 and 693.

Therefore, the required number is the H.C.F. of 441, 567 and 693 Respectively.

First, consider 441 and 567.

By applying Euclid's division lemma, we get

$567 = 441 \times 1 + 126$

$441 = 126 \times 3 + 63$

$126 = 63 \times 2 + 0.$

Therefore, H.C.F. of 441 and 567 = 63

Now, consider 63 and 693

again we have to apply Euclid's division lemma, we get

$693 = 63 \times 11 + 0.$

Therefore, H.C.F. of 441, 567 and 693 is 63

Hence, the required number is 63. 63 is the highest number which divides 445,572 and 699 will leave 4,5 and 6 as remainder respectively.

27. Let P and Q be the two positions of the plane and let A be the point of observation. Let ABC be the horizontal line through A. It is given that angles of elevation of the plane in two positions P and Q from point A are 60° and 30° respectively.

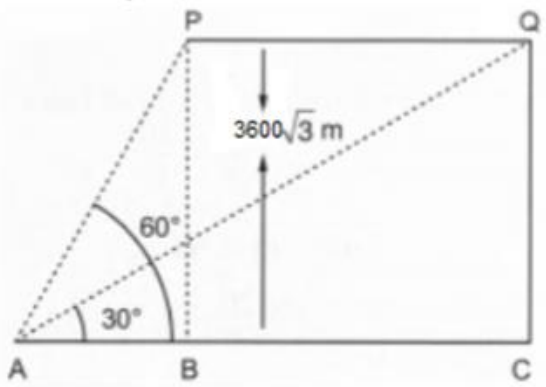
$\therefore \angle PAB = 60^\circ, \angle QAB = 30^\circ$. It is also given that $PB = 3600\sqrt{3} metres$

In $\triangle ABP$ we have:

$$\tan 60^\circ = \frac{BP}{AB}. \sqrt{3} = \frac{3600\sqrt{3}}{AB}.$$

$\Rightarrow AB = 3600\text{m}.$ In $\triangle ACQ$, we have:

$$\tan 30^\circ = \frac{CQ}{AC}.$$



$$\frac{1}{\sqrt{3}} = \frac{3600\sqrt{3}}{AC}.$$

$\Rightarrow AC = 3600 \times 3 = 10800\text{m}.$

$\therefore PQ = BC = AC - AB = 10800 - 3600 = 7200\text{m}.$

Thus, the plane travels 7200 m in 30 seconds.

$\text{Hence, Speed of plane} = \frac{7200}{30} = 240\text{m/sec}.$

$= \frac{240}{1000} \times 60 \times 60 = 864\text{km/hr}.$

28. Given, last term, l = 119

No. of terms in A.P. = 30

8th term from the end = 91

Let d be the common difference and assume that the first term of A.P. is 119 (from the end).

Since the n-th term of A.P. is:

$a_n = l + (n-1)d$

$\therefore a_8 = 119 + (8-1)d$

$\Rightarrow 91 = 119 + 7d$

$\Rightarrow 7d = 91 - 119$

$\Rightarrow 7d = -28$

$\Rightarrow d = -4$

Now, this common difference is from the end of the A.P.

So, the common difference from the beginning = -d

$= (-4) = 4$

Thus, a common difference for the A.P. is 4.

Now, using the formula:

$l = a + (n-1)d$

$\Rightarrow 119 = a + (30-1) \times 4$

$\Rightarrow 119 = a + 29 \times 4$

$\Rightarrow 119 = a + 116$

$\Rightarrow a = 119 - 116$

$\Rightarrow a = 3$

Hence, using the formula for the sum of n terms of an A.P.:

$S_n = \frac{n}{2}[2a + (n-1)d]$

$S_{30} = \frac{30}{2}[2 \times 3 + (30-1) \times 4]$

$= 15[6 + 29 \times 4]$

$= 15[6 + 116]$

$= 15 \times 122$

$= 1830$

Therefore, the sum of 30 terms of the A.P. is 1830.

OR

It is given that the sum of 5th and 9th terms of an A.P. is 72 and the sum of 7th and 12th terms is 97.

Let a be the first term and d be the common difference of the Arithmetic progression.

It is given that $a_5 + a_9 = 72$ and $a_7 + a_{12} = 97$.

$\Rightarrow (a+4d)+(a+8d)=72$ and $(a+6d)+(a+11d)=97.$

Therefore, we have:

$2a+12d=72 \quad \cdots(1)$

$2a+17d=97 \quad \cdots(2)$

Subtracting (1) from (2), we get:

$2a+17d-2a-12d=97-72$

$\Rightarrow 5d=25$

$\Rightarrow d=5$

Putting d = 5 in (1), we get:

$2a+12\times 5=72$

$\Rightarrow 2a+60=72$

$\Rightarrow 2a=12$

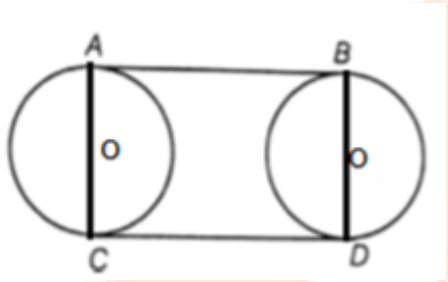
$\Rightarrow a=6$

Therefore, a = 6 and d = 5

Hence, the Arithmetic Progression $a, a+d, a+2d, \ldots$ is: $6, 11, 16, 21, 26, \ldots.$

29. Given: AB and CD are two common tangents to two circles of equal radii.

To prove



Construction: OA, OC, O'B and O'D

proof

Now, $\angle OAB = 90^\circ$ and $\angle OCD = 90^\circ$ as $OA \perp AB$ and $OC \perp CD$

A tangent at any point of a circle is perpendicular to the radius through the point of contact.

Thus, AC is a straight line.

Also, $\angle O'BA = \angle O'DC = 90^\circ$

A tangent at a point on the circle is perpendicular to the radius through the point of contact.

So, ABCD is a quadrilateral with four sides as AB, BC, CD and AD

But as $\angle A = \angle B = \angle C = \angle D = 90^\circ$

so, ABCD is a rectangle.

Hence, AB = CD (opposite sides of the rectangle are equal).

OR

$\therefore PT = PS$ (tangents from an external point P)

$\therefore \angle PTS = \angle PST$

Using Angle Sum Property in ΔPTS:

$\angle PTS + \angle PST + \angle TPS = 180^\circ \; 2\angle PTS = 180 - 60 = 120^\circ$

$\angle PTS = 60^\circ \Rightarrow PTS$ is an equilateral triangle.

So, TS = 4 cm.

Now, in $\triangle PTO : As PO is the angle bisector of \; \angle TPS, \; \angle OTP = 90^\circ$.

$$\tan 30^\circ = \frac{OT}{TP} \; \frac{1}{\sqrt{3}} = \frac{OT}{4}$$

$$OT = \frac{4}{\sqrt{3}} = \frac{4\sqrt{3}}{3} \text{cm.} \; \therefore \text{radius of circle} = \frac{4\sqrt{3}}{3} \text{cm.}$$

30. First, we will show that: $\frac{1+\tan^2 A}{1+\cot^2 A} = \tan^2 A$.

LHS: $\frac{1+\tan^2 A}{1+\cot^2 A} = \frac{1+\tan^2 A}{1+\frac{1}{\tan^2 A}} = \frac{1+\tan^2 A}{\frac{\tan^2 A+1}{\tan^2 A}} = (1+\tan^2 A) \times \frac{\tan^2 A}{\tan^2 A+1}.$

$= \tan^2 A = \text{RHS} \quad \cdots (i).$

Now, we will show that:

$$\left(\frac{1-\tan A}{1-\cot A}\right)^2 = \tan^2 A.$$

LHS:

$$\left(\frac{1-\tan A}{1-\cot A}\right)^2 = \left(\frac{1-\tan A}{1-\frac{1}{\tan A}}\right)^2 = \left(\frac{1-\tan A}{\frac{\tan A-1}{\tan A}}\right)^2 = \left[(1-\tan A)\times\frac{\tan A}{\tan A-1}\right]^2.$$

$$= \left(\frac{-\tan A(1-\tan A)}{-(\tan A-1)}\right)^2 = (\tan A)^2 = \tan^2 A = \text{RHS} \quad \cdots(ii).$$

Hence, from (i) and (ii):

$$\frac{1+\tan^2 A}{1+\cot^2 A} = \left(\frac{1-\tan A}{1-\cot A}\right)^2 = \tan^2 A.$$

Hence proved.

31.

Class Interval	**Frequency (f_i)**	**Midpoint (x_i)**	$f_i \cdot x_i$
0-10	12	5	60
10-20	18	15	270
20-30	27	25	675
30-40	20	35	700
40-50	17	45	765

50-60	6	55	330
Total	**100**		**2800**

$$\text{Mean} = \frac{\sum f_i x_i}{\sum f_i} = \frac{2800}{100} = 28$$

Section D

32. Let the cost of production of each article be Rs x.

We are given that the total cost of production on that particular day is Rs 90.

Therefore, the total number of articles produced that day = $\frac{90}{x}$.

According to the given conditions:

$$x = 2\left(\frac{90}{x}\right) + 3$$

$$\Rightarrow x = \frac{180}{x} + 3$$

$$\Rightarrow x^2 = 180 + 3x$$

$$\Rightarrow x^2 - 3x - 180 = 0$$

$$\Rightarrow x^2 - 15x + 12x - 180 = 0$$

$$\Rightarrow x(x-15) + 12(x-15) = 0$$

$$\Rightarrow (x-15)(x+12) = 0 \quad \Rightarrow x = 15, x = -12$$

Cost cannot be negative, therefore, we discard x = -12.

Therefore, x = Rs 15, which is the cost of production of each article.

Number of articles produced on that particular day:

$= \frac{90}{15} = 6.$

OR

Since -5 is a root of the given quadratic equation $2x^2 + px + 15 = 0$, then:

$2(-5)^2 + p(-5) + 15 = 0$

$50 - 5p + 15 = 0$

$\Rightarrow 5p = 35 \quad \Rightarrow p = 7.$

Now $p(x^2 + x) + k = 0$ has equal roots:

$px^2 + px + k = 0$

So, $(b^2) - 4ac = 0$

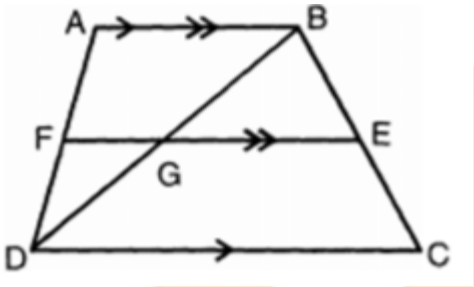
$(7)^2 - 4ac = 0$

$49 - 4 \times 7 \times k = 0$

$28k = 49$

$k = \frac{49}{28} = \frac{7}{4}.$

33.



In a trapezium ABCD, $AB \| DC$, $EF \| AB$, and $CD = 2AB$

It is also given that:

$\frac{BE}{EC} = \frac{4}{3} \quad \cdots(1)$

Since $AB \| CD$ and $AB \| EF$,

$\frac{AF}{FD} = \frac{BE}{EC} = \frac{4}{3}.$

In ΔBGE and ΔBDC:

$\angle BEG = \angle BCD$ (corresponding angles),

$\angle GBE = \angle DBC$ (common).

$\therefore \Delta BGE \sim \Delta BDC$ (by AA similarity).

$\Rightarrow \frac{EG}{CD} = \frac{BE}{BC} \quad \cdots(2).$

Now, from (1):

$\frac{EC}{BC} = \frac{3}{4}.$

$\Rightarrow \frac{BE}{BC} = 1 - \frac{EC}{BC} = 1 - \frac{3}{4} = \frac{1}{4}.$

From equation (2):

$\frac{EG}{CD} = \frac{BE}{BC} = \frac{4}{7}.$

$\therefore EG = \frac{4}{7}CD \quad \cdots(3).$

Similarly, $\Delta DGF \sim \Delta DBA$(by AA similarity).

Similarly, $\Delta DGF \sim \Delta DBA$ (by AA similarity):

$\frac{DE}{DA} = \frac{FG}{AB}.$

$\Rightarrow \frac{FG}{AB} = \frac{3}{7}.$

$\Rightarrow FG = \frac{3}{7}AB \quad \cdots(4).$

Since:

$\frac{AF}{AD} = \frac{4}{7}, \quad \frac{BE}{BC} = \frac{3}{7} = \frac{DE}{DA},$

adding equations (3) and (4), we get:

$EG + FG = \frac{4}{7}CD + \frac{3}{7}AB.$

$\Rightarrow EF = \frac{4}{7} \times (2AB) + \frac{3}{7}AB.$

$\Rightarrow EF = \frac{8}{7}AB + \frac{3}{7}AB = \frac{11}{7}AB.$

$\therefore 7EF = 11AB.$

34. Given, radius of cone = radius of hemisphere: $r = 7\text{cm}.$

Height of cone (h) = $2 \times$ radius :

$h = 2 \times 7 = 14\text{cm}.$

Volume of solid = Volume of cone + Volume of hemisphere:

Volume of solid $(V) = \frac{1}{3}\pi r^2 h + \frac{2}{3}\pi r^3.$

Substituting h = 2r:

$V = \frac{1}{3}\pi r^2 (2r) + \frac{2}{3}\pi r^3.$

$V = \frac{2}{3}\pi r^3 + \frac{2}{3}\pi r^3.$

$V = \frac{4}{3}\pi r^3.$

Substituting r = 7 cm:

$V = \frac{4}{3} \times \frac{22}{7} \times 7 \times 7 \times 7.$

$V = \frac{4312}{3}\text{cm}^3.$

$V = 1437.33\text{cm}^3.$

OR

Height of the cylinder = 3 m.

Total height of the tent above the ground = 13.5 m.

Height of the cone = (13.5 - 3) = 10.5 m.

Radius of the cylinder = radius of the cone = 14 m.

Curved surface area of the cylinder = $2\pi rh = 2\times\frac{22}{7}\times14\times3 = 264m^2$.

$l = \sqrt{r^2+h^2} = \sqrt{14^2+(10.5)^2} = \sqrt{196+110.25} = \sqrt{306.25} = 17.5$.

Curved surface area of the cone $= \pi rl = \frac{22}{7}\times14\times17.5 = 770m^2$.

Let S be the total area which is to be painted. Then,

S = Curved surface area of the cylinder + Curved surface area of the cone

$\Rightarrow S = (264+770)m^2 = 1034m^2$.

Hence, Cost of painting = $S\times\text{Rate} = 1034\times2 = 2068$ Rupees.

35.

Class Interval	**Frequency (f_i)**	**Class Marks (x_i)**	$u_i = \frac{x_i - A}{h}$	$f_i \cdot u_i$
5.5 - 15.5	6	10.5	-3	-18

15.5 - 25.5	11	20.5	-2	-22
25.5 - 35.5	21	30.5	-1	-21
35.5 - 45.5	23	40.5	0	0
45.5 - 55.5	14	50.5	1	14
55.5 - 65.5	5	60.5	2	10
Total	**80**			**-37**

A = 40.5, h = 10

we know that, Mean $\bar{x} = \text{A} + \dfrac{h \times \sum f_i u_i}{\sum f_i}$

$= 40.5 + \dfrac{10 \times (-37)}{80}$

$= 40.5 + \dfrac{-370}{80}$

= 40.5 - 4.63 = 35.87

The modal class of the given data is 35.5 - 45.5.

$\therefore x_k = 35.5$, h = 10, $f_k = 23$, $f_{k-1} = 21$, $f_{k+1} = 14$

we know that, Mode, $M_0 = x_k + \dfrac{h \times (f_k - f_{k-1})}{(2f_k - f_{k-1} - f_{k+1})}$

$= 35.5 + \dfrac{10 \times (23-21)}{(2 \times 23 - 21 - 14)}$

$= 35.5 + \frac{10 \times 2}{11}$

$= 35.5 + 1.82 = 37.32$

Clearly, mode > mean

Section E

36. i. Given, prize amount for Hockey ₹ x and ₹ y for cricket per student

∴ Algebraic equations are $5x + 4y = 9500$ ...(i)

and $4x + 3y = 7370$ ...(ii)

ii. Given, prize amount for Hockey ₹ x and ₹ y for cricket per student

∴ Algebraic equations are

$5x + 4y = 9500$ ...(i)

and $4x + 3y = 7370$ ...(ii)

Multiply by 3 in equation (i) and by 4 in equation (ii)

$15x + 12y = 28,500$ ...(iii)

$16x + 12y = 29480$ ...(iv)

On subtracting equation (iii) from equation (iv), we get

$x = 980$

∴ Prize amount for hockey = ₹ 980

iii. Given, prize amount for Hockey ₹ x and ₹ y for cricket per student

∴ Algebraic equations are 5x + 4y = 9500 ...(i)

and 4x + 3y = 7370 ...(ii)

Now, put this value in equation (i), we get

5 × 980 + 4y = 9500

⇒ 4y = 9500 - 4900 = 4600

⇒ y = 1150

∴ Prize amount for cricket = ₹ 1150

Difference = 1150 - 980 = 170

∴ Prize amount for cricket is ₹ 170 more than hockey.

OR

Total prize amount for 2 students each from two games

= 2x + 2y

= 2(x + y)

= 2(980 + 1150)

= 2 × 2130

= ₹ 4260

37. i. Point of intersection of graph of polynomial, gives the zeroes of the polynomial.

∴ zeroes = -4 and 7

ii. Since, zero's are $\alpha = -4$, $\beta = 7$

$\alpha+\beta = -4 + 7 = 3$

$\alpha\beta = -4 \times 7 = -28$

$P(x) = x^2$ - (Sum of zeroes)x + product of zeroes

$P(x) = x^2 - 3x + (-28)$

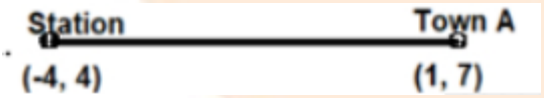
$P(x) = x^2 - 3x - 28$

iii. Product of zeroes = $-4 \times 7 = -28$

OR

a is a non-zero real number, b and c are any real numbers c.

38. i.



Distance travelled by Veena = $\sqrt{(1-(-4))^2+(7-4)^2}$

$= \sqrt{5^2+3^2}$

$= \sqrt{25+9}$

$= \sqrt{34}$



Distance travelled by Arun $= \sqrt{(4-(-4))^2 + (2-4)^2}$

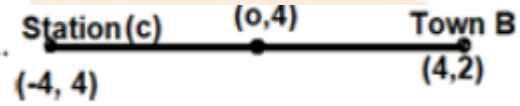
$= \sqrt{64+4}$

$= \sqrt{68}$

Therefore, Arun will travel more distance to reach his home.

ii. Location of station = (-4, 4)

iii.

i. 

Let y-axis divides station (c) and Town B in K : 1

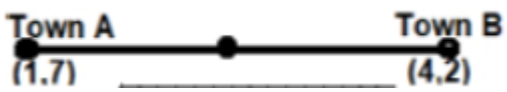
$0 = \frac{4k-4}{k+1}$

4k = 4

k = 1

∴ y-axis divides in 1 : 1

OR



$AB = \sqrt{(4-1)^2 + (2-7)^2}$

$= \sqrt{9+25}$

$= \sqrt{34}$